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EDITED BY
Mustafa Kurt,
Near East University, Cyprus

REVIEWED BY
Noble Lo,
Lancaster University, United Kingdom
Eko Indrawan,
Padang State University, Indonesia

*CORRESPONDENCE
Dong Han
✉ handong@nenu.edu.cn

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A study on the stimulation and guidance of intrinsic motivation among engineering college students in China

Kunyue Yang and Dong Han*

School of Information Science and Technology, Northeast Normal University, Changchun, China

Background: Against the backdrop of rapid advancements in artificial intelligence and information technology, engineering university students face complex learning challenges. However, existing tools for measuring their intrinsic motivation—a crucial driver of learning—often lack cultural adaptability in the Chinese context, and systematic intervention strategies are scarce.
Methods: Based on Self-Determination Theory and the Intrinsic Motivation Inventory, we developed and locally adapted the Chinese College Student's Intrinsic Motivation Learning Questionnaire (CIMLQ), removing the less applicable "Relatedness" dimension. The final 30-item instrument covers six dimensions: Interest/Enjoyment, Perceived Competence, Effort, Pressure/Tension, Perceived Choice, and Value/Usefulness. Its reliability and validity were tested with 450 engineering students. Subsequently, a 19-week randomized controlled trial (N = 600) was conducted, comparing an experimental group receiving multidimensional motivation strategies (e.g., goal contracting, choice provision) with a control group under conventional instruction.
Results: Exploratory and confirmatory factor analyses demonstrated good structural validity (cumulative variance explained: 66.13%) and internal consistency (Cronbach's α = 0.829–0.891). The teaching experiment revealed that the experimental group showed a significantly higher post-intervention intrinsic motivation score (119.20 vs. 97.81, $*p* < 0.001$, a 21.8% increase) and better academic performance (80.10 vs. 76.57, $*p* = 0.001$) compared to the control group.
Conclusion: The CIMLQ proves to be a reliable and valid measure for assessing intrinsic motivation among Chinese engineering students. The findings provide empirical evidence that structured instructional interventions, designed based on the CIMLQ dimensions, can effectively enhance both intrinsic motivation and academic outcomes. This study offers a practical framework for fostering self-directed learning capabilities that are essential for adapting to evolving educational environments.

KEYWORDS

college student, intervention, intrinsic motivation, measurement, testing

1 Introduction

Against the backdrop of the rapidly evolving information society and the advent of the artificial intelligence era, talents in information science have become a crucial force driving contemporary development. However, due to the inherent complexity and high difficulty of disciplines within information science and technology, engineering university students face particularly pronounced learning challenges. Research has identified key issues in their learning processes, including a lack of proactive learning, poor self-management, low classroom engagement, and deficiencies in autonomous planning and self-discipline.

Consequently, effectively stimulating and sustainably guiding university students' intrinsic motivation for self-regulated learning has emerged as a significant research topic in the field of education.

Intrinsic motivation, as the core driver of self-regulated learning, directly determines an individual's willingness to actively engage, persistently explore, and pursue deep understanding. As emphasized by Self-Determination Theory (SDT), intrinsic motivation is the cornerstone of autonomous learning—deep, sustainable self-regulated learning occurs only when it genuinely originates from an inner desire to learn ("I want to learn") rather than being controlled by external pressure (Howard et al., 2021). "Self-regulated learning" lacking the support of intrinsic motivation often becomes superficial (Taylor et al., 2014).

Furthermore, the cultivation and stimulation of intrinsic motivation hold multi-faceted significance for the holistic development and academic success of contemporary university students. At the cognitive level, it drives deep learning and the development of higher-order thinking skills (Taylor et al., 2014). At the behavioral level, it significantly enhances self-regulated learning capabilities and academic persistence (Järvelä and Bannert, 2021). At the psychological level, it strengthens self-efficacy and resilience in coping with academic stress (Reeve and Lee, 2014). Thus, cultivating intrinsic motivation is crucial for developing self-directed learners who can effectively plan, monitor, and reflect on their learning—skills that are increasingly valuable in complex and technology-rich learning contexts.

Intrinsic motivation refers to the behavioral drive spontaneously generated by an individual due to intrinsic interest or curiosity (Ryan and Deci, 2000). This motivation stems from a natural inclination and enthusiasm for specific activities or goals, rather than external rewards or pressure. Its theoretical foundation can be traced back to Self-Determination Theory (SDT). Formally proposed by Deci and Ryan (1985), this theory conceptualizes motivation as a continuum ranging from amotivation to external motivation and finally to intrinsic motivation. Ryan and Deci (2017) systematically elaborated that intrinsic motivation arises from the spontaneous pursuit of an activity's inherent value, and the degree to which the three basic psychological needs—autonomy, competence, and relatedness—are satisfied directly determines the quality of an individual's motivation and psychological well-being. This theoretical framework, supported by early experiments demonstrating the "undermining effect" of external rewards on intrinsic motivation and the positive role of autonomy-supportive environments, provides core theoretical support for subsequent educational research.

The development of scales to measure intrinsic motivation has undergone key evolution. The Academic Motivation Scale (AMS) developed by Vallerand et al. (1992) laid the foundation for measuring the motivational regulation continuum. The recent proliferation of blended learning has propelled questionnaire validation into a new phase, with numerous studies verifying the construct validity of intrinsic motivation in online learning contexts (Hartnett, 2016). For instance, Tsai et al. (2018) study on learners in Massive Open Online Courses (MOOCs) indicated that learners' metacognitive abilities, by enhancing intrinsic motivational elements like interest/enjoyment, significantly predicted their continuance intention.

Intrinsic motivation and self-regulated learning are increasingly understood as culturally patterned and feedback-sensitive constructs. A qualitative comparison between Hong Kong and UK students reveals distinct motivational emphases: Hong Kong students often exhibit stronger orientation toward achievement and competition, whereas UK students prioritize personal development and subject passion, as identified in cross-cultural comparative research (Lo et al., 2024). These profiles correlate with differing independent learning approaches, with students from Hong Kong and the UK demonstrating differing conceptualizations and implementations of independent learning strategies, as identified in cross-cultural comparative research (Lo et al., 2024). Importantly, the pedagogical design of feedback—a core element in supporting basic psychological needs—can modulate these motivational pathways. Evidence suggests that low-friction feedback cycles (e.g., brief prompts, visual progress checks) can enhance perceived competence and autonomy without elevating pressure (Lo et al., 2025). This integrated perspective justifies a culturally responsive approach, supporting our adaptation of the CIMLQ and informing our intervention design that employs structured yet minimally intrusive feedback mechanisms to target autonomy, competence, and value within a Chinese engineering context.

Although the theoretical and applied value of intrinsic motivation questionnaires is widely recognized, their application in higher education practice still faces three main gaps, which constitute the core proposition this study aims to address:

(1) Limitations of Measurement Tools: The structural validity of existing intrinsic motivation scales (e.g., IMI) faces challenges in cross-cultural applications, with factor structures often showing instability in non-Western samples. Directly transplanting Western psychometric tools can lead to bias, highlighting the necessity for localized validation (Chen et al., 2005).
(2) Oversimplified Examination of Cultivation Strategies: As noted by Patall et al. (2021), current intervention research often examines single strategies in isolation (e.g., autonomy support or feedback), lacking comparisons of the effectiveness of different strategy combinations within the same educational context.
(3) Scarcity of Empirical Research in Authentic Educational Settings: A systematic review of meta-analyses by Schneider and Preckel (2017) pointed out that long-term, classroom-based longitudinal studies remain scarce in higher education motivation research, particularly regarding stimulating and guiding intrinsic motivation in university students.

Furthermore, recent cross-cultural scholarship provides complementary perspectives to contextualize this study. First, feedback is not a neutral component of instructional design; its modality and immediacy can shape students' motivation, perception of guidance quality, and performance (Yang and Chan, 2023). Second, robust cultural patterns exist in independent learning strategies and motivational profiles. Comparative research indicates systematic differences in how students from Hong Kong and the UK conceptualize and enact independent learning strategies, reflecting distinct educational cultural contexts (Lo et al., 2024). These insights underscore the necessity for culturally attuned measurement and intervention. The present study addresses these gaps by developing a localized instrument (CIMLQ) and testing an intervention that incorporates structured,

low-friction feedback cycles (e.g., goal cards, reflection prompts) designed to leverage key psychological needs within the specific context of Chinese engineering education.

2 Materials and methods

2.1 Participants and procedure

This study was conducted in two sequential phases: a scale development and validation phase, followed by an intervention experiment. For the initial scale validation, 500 undergraduate students from the School of Information Science and Technology at Northeast Normal University were recruited. After data cleaning, 450 valid questionnaires were retained for psychometric analysis. For the subsequent intervention experiment, a distinct cohort of 600 undergraduates from the same school was randomly selected at the beginning of the Fall 2024 semester.

A total of 650 students were assessed for eligibility for the intervention experiment. Of these, 600 met the criteria and provided informed consent, constituting the final cohort. They were randomly assigned at the individual level (1:1) to an experimental group (n = 300) or a control group (n = 300) using computer-based randomization. Allocation concealment was ensured as group assignment was concealed from the instructors until the start of the intervention. Thereafter, due to the nature of the educational intervention, blinding of the instructors (who delivered the intervention) and participants was not feasible, but data analysts were blinded to group allocation. To minimize contamination, the two groups attended separate course sessions. All randomized participants (n = 600) completed the study, with no attrition. The participant flow is detailed in Figure 1. This study was conducted in accordance with the Declaration of Helsinki.

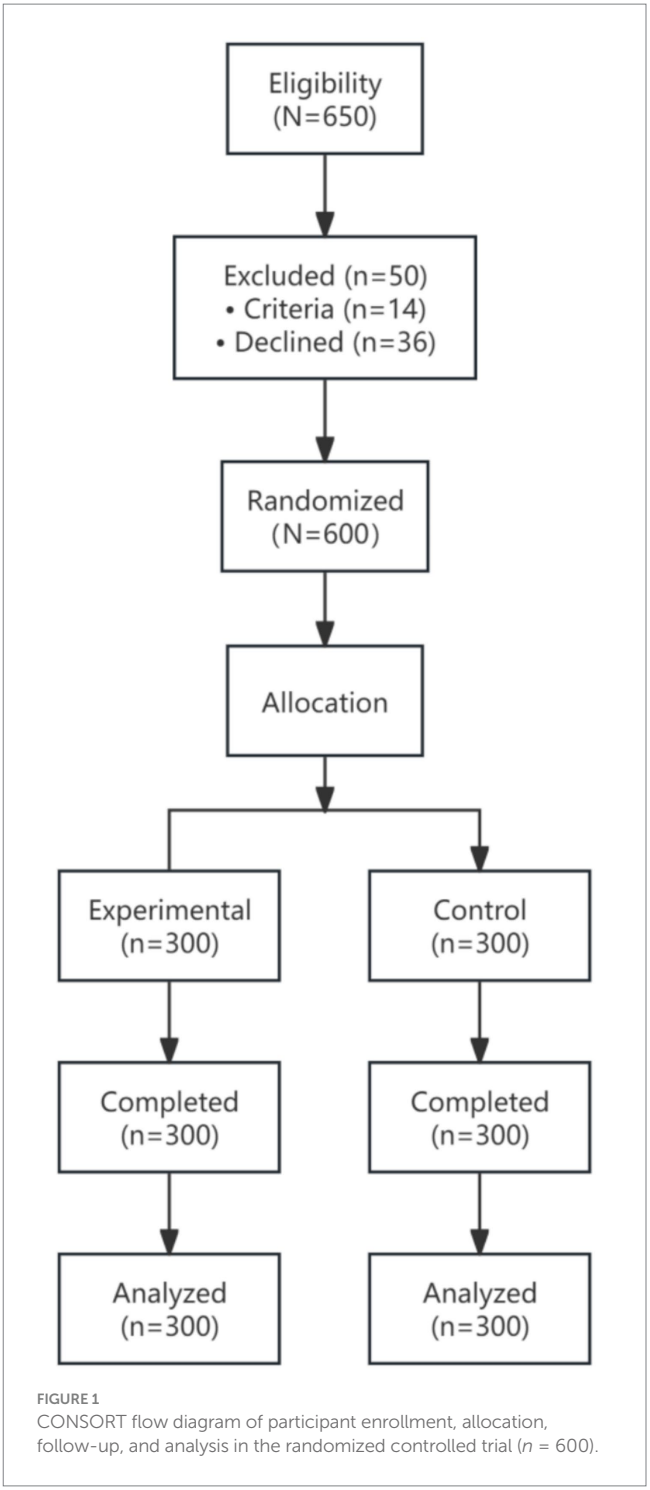


FIGURE 1
CONSORT flow diagram of participant enrollment, allocation, follow-up, and analysis in the randomized controlled trial (n = 600).

2.2 CIMLQ development and cultural adaptation

The Chinese College Student Intrinsic Motivation Learning Questionnaire (CIMLQ) was developed through a systematic cultural adaptation of the Intrinsic Motivation Inventory (IMI), guided by Self-Determination Theory.

Adaptation Process. The process followed established cross-cultural adaptation guidelines and comprised four stages: (1) independent forward translation and synthesis by bilingual researchers; (2) review by an expert panel (n = 3) for semantic, conceptual, and cultural relevance; (3) cognitive interviews with target students (n = 8) to refine item clarity; and (4) a pilot test (n = 50) to confirm item adequacy.

Dimension Selection. The 'Relatedness' dimension was excluded based on two key considerations. First, psychometric evidence from prior adaptations in Chinese contexts indicates its unstable performance (e.g., Li and Wang, 2022). Second, culturally, the need for connection in Chinese classrooms may manifest in ways not fully captured by the original IMI items. This exclusion is a measurement optimization for this study, not a theoretical claim. The final CIMLQ thus assesses six dimensions: Interest/Enjoyment, Perceived Competence, Effort, Pressure/Tension, Perceived Choice, and Value/Usefulness.

Final Instrument. The CIMLQ is a 30-item self-report measure (5 items per dimension, each including one reverse-scored item), rated on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). The complete item list, scoring instructions, and both Chinese and English versions are provided in the Supplementary materials ("Chinese College Student Intrinsic Motivation Learning Questionnaire (CIMLQ) Full Item List and Scoring Instructions").

2.3 Intervention design

A 19-week randomized controlled trial was implemented within the natural teaching framework of the Fall 2024 semester. The experimental group received a systematic, multidimensional intervention designed to stimulate intrinsic motivation, which was embedded into regular course teaching.

The intervention was delivered by the students' regular course instructors. Prior to the semester, these instructors attended a single 3-h briefing session that outlined the intervention rationale and procedures, and they were provided with a guidebook containing examples of goal cards and reflective prompts. To ensure adherence to the protocol, the research team reviewed a random sample of 20% of the submitted student materials (goal cards and journals) each month.

The specific, theory-aligned strategies were as follows:

1. Goal Commitment Reinforcement (supporting Autonomy). To enhance learning initiative and perceived choice, students created weekly learning goal cards. Teachers used brief (approximately 5-min) open-ended prompts at the start or end of selected weekly sessions to facilitate reflection on these goals.
2. Cognitive Reflection & Group Sharing (supporting Relatedness). To normalize challenges and foster a supportive learning community, students were encouraged to write short reflective journals following major assignments or exams (approximately 2–3 times per semester). Subsequently, they discussed their experiences and coping strategies in small-group discussions (6–8 students per group) integrated into regular tutorial sessions once per month.
3. Visual Progress Monitoring (supporting Competence). To reinforce self-management and provide tangible evidence of progress, students used a simple checklist to mark the completion of core weekly tasks. They were guided to briefly review their weekly completion rate during the goal reflection time.

The control group strictly maintained conventional teaching and management modes and received no experimental intervention. The CIMLQ was administered as a pre-test at the semester start and a post-test at the semester end to all 600 participants. Their final exam scores were also recorded for comparative analysis with baseline academic performance.

Academic performance was assessed using a common final examination administered identically to both groups at the end of the semester. Scoring was conducted by course instructors who were blinded to participants' group allocation.

2.4 Data analysis plan

Data analysis was performed using SPSS 26.0 for reliability testing and intervention analysis, and AMOS 24.0 for confirmatory factor analysis (CFA). The validation of the CIMLQ involved assessing internal consistency using Cronbach's alpha coefficient and examining structural validity through both Exploratory Factor Analysis (EFA) and CFA. To test the intervention effects, both between-groups and within-groups comparisons were employed. Analysis of variance (ANOVA) was used to compare post-intervention scores between the control and experimental groups. Paired-sample t-tests were used to compare pre-test and post-test data within each group. An independent samples t-test was used to compare the final academic performance (exam scores) between the two groups.

No missing data were present in this study; all 600 participants provided complete questionnaire responses and exam scores at both time points. Listwise deletion was applied, and only complete cases were analyzed.

3 Results

3.1 Psychometric properties of the CIMLQ

The Exploratory Factor Analysis (EFA) for the CIMLQ, based on the validation sample ($N = 450$), indicated that the data were suitable for factor analysis, with a Kaiser-Meyer-Olkin (KMO) measure of 0.869. The six-factor model explained a cumulative variance of 66.13%. The factor loadings for all 30 items are presented in Table 1.

The internal consistency of the CIMLQ was excellent. The Cronbach's alpha coefficients for all six dimensions were significantly above the threshold of 0.70, ranging from 0.835 to 0.891, and the total questionnaire's alpha reached 0.829. The complete reliability coefficients are shown in Table 2.

To validate the factor structure of the full instrument, a confirmatory factor analysis (CFA) was conducted on the complete 30-item, six-factor model. The model demonstrated excellent fit to the data, with all key fit indices meeting or exceeding recommended thresholds: $\chi^2/df = 1.918$, CFI = 0.944, TLI = 0.935, RMSEA = 0.045 (90% CI [0.040, 0.050]), and SRMR = 0.043 (see Table 3).

To confirm the superiority of the hypothesized six-factor structure, we compared it against two theoretically plausible alternative models: a one-factor model (all items loading on a single "intrinsic motivation" factor) and a two-factor model (items grouped into positive and negative valence factors). As shown in Table 4, the six-factor model provided a significantly better fit to the data than both alternative models, supporting the discriminant validity of the six dimensions.

Convergent validity was further assessed by calculating the average variance extracted (AVE) and composite reliability (CR) for each factor. As reported in Table 5, all six factors demonstrated strong convergent validity, with AVE values exceeding 0.50 (range: 0.510–0.624) and CR values above 0.70 (range: 0.837–0.892). Standardized factor loadings for all 30 items are provided in Supplementary material S1 (range: 0.671–0.840).

Additionally, separate CFA models were tested for each of the six constructs individually. All single-construct models also demonstrated acceptable fit, with detailed indices provided in Table 6.

Furthermore, the correlations between the tested dimensions were all significantly different from unity, providing strong evidence for the discriminant validity of the model. The results of the discriminant validity tests are summarized in Table 7.

Discriminant validity was evaluated using two complementary approaches. First, the Fornell-Larcker criterion was applied, whereby the square root of the average variance extracted ($\sqrt{\text{AVE}}$) for each construct should be greater than its correlations with any other construct. As presented in Table 8, the $\sqrt{\text{AVE}}$ for each factor (values on the diagonal) exceeded all corresponding inter-construct

TABLE 1 Factor analysis results for the CIMLQ (n = 450).

Item	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6
a-1	**0.762**	–	–	–	–	–
a-2	**0.774**	–	–	–	–	–
a-3	**0.800**	–	–	–	–	–
a-4	**0.770**	–	–	–	–	–
a-5	**0.777**	–	–	–	–	–
b-1	–	**0.749**	–	–	–	–
b-3	–	**0.793**	–	–	–	–
b-4	–	**0.798**	–	–	–	–
b-5	–	**0.777**	–	–	–	–
c-1	–	–	**0.799**	–	–	–
c-2	–	–	**0.767**	–	–	–
c-3	–	–	**0.774**	–	–	–
c-4	–	–	**0.793**	–	–	–
c-5	–	–	**0.741**	–	–	–
d-1	–	–	–	**0.824**	–	–
d-2	–	–	–	**0.808**	–	–
d-3	–	–	–	**0.789**	–	–
d-4	–	–	–	**0.801**	–	–
d-5	–	–	–	**0.828**	–	–
e-1	–	–	–	–	**0.780**	–
e-2	–	–	–	–	**0.832**	–
e-3	–	–	–	–	**0.810**	–
e-4	–	–	–	–	**0.762**	–
e-5	–	–	–	–	**0.832**	–
f-1	–	–	–	–	–	**0.811**
f-2	–	–	–	–	–	**0.796**
f-3	–	–	–	–	–	**0.792**
f-4	–	–	–	–	–	**0.816**
f-5	–	–	–	–	–	**0.840**
Eigenvalue	3.055	3.089	3.044	3.587	3.508	3.557
% Variance	10.184	10.295	10.146	11.955	11.692	11.858
Cumulative Variance %	66.130	–	–	–	–	–

Bold values indicate factor loadings ≥ 0.700.

TABLE 2 Cronbach's alpha coefficients for the CIMLQ (n = 450).

Construct	Interest/ enjoyment	Perceived competence	Effort	Pressure/ tension	Perceived choice	Value/ usefulness	Total questionnaire
No. of Items	5	5	5	5	5	5	30
α Coefficient	0.891	0.841	0.836	0.835	0.885	0.889	0.829

correlations (off-diagonal values), meeting this criterion. Second, we computed the heterotrait-monotrait (HTMT) ratio of correlations, a more robust measure. As shown in Table 9, all HTMT values were well below the recommended threshold of 0.85, providing strong evidence that the six dimensions are empirically distinct.

Potential method bias due to reverse-worded items was assessed. Given that each dimension contains one reverse-scored item, we tested whether a common method factor model (specifying correlated residuals among all six reverse-worded items) would yield a significantly better fit than the baseline six-factor model. The comparison of model fit indices is detailed in Supplementary Table S2. The results indicated that the method-factor model did not provide a statistically significant improvement in fit ($\Delta\chi^2(15) = 17.209$, $p > 0.05$; ΔCFI = 0.001). Therefore, substantial method bias attributable to reverse wording was not detected in this sample, and the baseline model was retained for all subsequent analyses.

Assumptions for parametric tests were checked. Shapiro–Wilk tests indicated normal distribution for total scores in all groups at both

time points (all $p > 0.05$). Homogeneity of variance was violated at post-test between experimental and control groups (Levene's $F = 41.58$, $p < 0.001$), consistent with the intervention's effect of creating more homogeneous performance. We therefore report Welch's t-test results where equal variances are not assumed.

3.2 Intervention effects on intrinsic motivation

Baseline equivalence was assessed using an independent-samples *t*-test on pre-test intrinsic motivation total scores. No significant difference was found between groups, *t*(598) = 0.97, *p* = 0.335, Cohen's *d* = 0.08 [−0.08, 0.24], indicating successful randomization. This single comparison replaces the original 30 item-level tests to avoid Type I error inflation.

To examine the intervention effect on intrinsic motivation, we fitted a linear mixed-effects model with a random intercept for student ID and fixed effects for Group, Time, and their interaction. The analysis revealed a significant Group × Time interaction, $F(1, 598) = 510.12$, $*p* < 0.001$, partial $\eta^2 = 0.460$, indicating that the change in motivation over time differed between groups. Post-hoc comparisons showed that the experimental group's total intrinsic motivation score increased significantly from pre- to post-test (Cohen's *d* = 1.87), whereas the control group's score remained stable (Cohen's *d* = −0.01). At the post-test, the experimental group scored significantly higher than the control group, *t*(598) = 21.34, $*p* < 0.001$, Cohen's *d* = 1.74. Complete results are presented in Table 10.

The within-group changes are summarized in the lower section of Table 10. For the experimental group, the increase in total intrinsic motivation from pre- to post-test was statistically significant and large in magnitude (Cohen's *d* = 1.87). In contrast, the control group showed no significant change over the same period (Cohen's *d* = −0.01). This pattern confirms that the observed growth in motivation was specific to the intervention condition.

TABLE 3 Overall fit indices for the six-factor CFA model ($N = 450$).

Fit Index	Value	Interpretation
χ^2/df	1.918	Excellent (< 3.0)
CFI	0.944	Excellent (> 0.90)
TLI	0.935	Excellent (> 0.90)
RMSEA (90% CI)	0.045 [0.040, 0.050]	Excellent (< 0.08)
SRMR	0.043	Excellent (< 0.08)

CFA, confirmatory factor analysis; CI, confidence interval.

TABLE 4 Model fit comparisons for alternative CFA models ($N = 450$).

Model	χ^2	df	χ^2/df	CFI	TLI	RMSEA	SRMR
Six-factor	721.32	376	1.918	0.944	0.935	0.045	0.043
One-factor	4847.92	406	11.941	0.277	0.225	0.156	0.343
Two-factor (Positive/ Negative)	3961.01	406	9.756	0.421	0.380	0.140	0.279

TABLE 5 Convergent validity indicators for the six-factor CIMLQ model ($N = 450$).

Construct	Items	Standardized loadings range	CR	AVE	McDonald's ω
Interest/Enjoyment	5	0.714–0.797	0.864	0.559	0.861
Perceived Competence	5	0.671–0.741	0.841	0.516	0.835
Effort	5	0.683–0.744	0.837	0.510	0.833
Pressure/Tension	5	0.740–0.814	0.892	0.624	0.897
Perceived Choice	5	0.729–0.817	0.886	0.608	0.889
Value/Usefulness	5	0.756–0.808	0.889	0.617	0.895

CR, composite reliability; AVE, average variance extracted. All AVE > 0.50 and CR > 0.70, supporting convergent validity.

TABLE 6 Convergent validity of the questionnaire model ($n = 450$).

Construct	df	p	χ^2/df	GFI	AGFI	NFI	RFI	IFI	TLI	CFI	RMSEA
Perceived Choice	5	0.341	1.131	0.995	0.985	0.993	0.998	0.999	0.998	0.999	0.017
Pressure/Tension	5	0.353	1.110	0.995	0.986	0.993	0.999	0.999	0.999	0.999	0.016
Value/Usefulness	5	0.111	1.789	0.992	0.976	0.989	0.990	0.995	0.990	0.995	0.042
Interest/Enjoyment	5	0.526	0.833	0.996	0.989	0.997	0.999	0.999	0.999	0.999	0.012
Effort	5	0.771	0.508	0.998	0.993	0.998	0.999	0.999	0.999	0.999	0.024
Perceived Competence	5	0.284	1.248	0.994	0.983	0.995	0.998	0.999	0.998	0.998	0.023

TABLE 7 Discriminant validity of the model dimensions (n = 450).

Model	χ^2	df	$\Delta\chi^2$
1. Unconstrained Model	150.00	50	–
2. Correlation between Perceived Choice & Pressure/Tension fixed to 1	180.50	51	30.500***
3. Correlation between Perceived Choice & Value/Usefulness fixed to 1	190.00	51	40.000***
4. Correlation between Perceived Choice & Interest/Enjoyment fixed to 1	200.00	51	50.000***
5. Correlation between Perceived Choice & Effort fixed to 1	210.00	51	60.000***
6. Correlation between Perceived Choice & Perceived Competence fixed to 1	220.00	51	70.000***
7. Correlation between Pressure/Tension & Value/Usefulness fixed to 1	180.00	51	30.000**
8. Correlation between Pressure/Tension & Interest/Enjoyment fixed to 1	190.50	51	40.500***
9. Correlation between Pressure/Tension & Effort fixed to 1	200.50	51	50.500***
10. Correlation between Pressure/Tension & Perceived Competence fixed to 1	210.50	51	60.500***

***$p < 0.001$, *$p < 0.01$.

TABLE 8 Discriminant validity assessment via the Fornell-Larcker criterion (N = 450).

Construct	(1)	(2)	(3)	(4)	(5)	(6)
(1) Interest/Enjoyment	**0.748**					
(2) Perceived Competence	0.003	**0.718**				
(3) Effort	0.068	0.069	**0.714**			
(4) Pressure/Tension	0.051	0.026	0.168	**0.790**		
(5) Perceived Choice	0.044	0.031	0.147	0.412	**0.779**	
(6) Value/Usefulness	0.131	0.030	0.115	0.370	0.361	**0.786**

Diagonal elements (in bold) are the square roots of the Average Variance Extracted (AVE) for each construct. Off-diagonal elements are the correlations between constructs. Discriminant validity is established when the diagonal value ($\sqrt{}$AVE) is greater than the correlations in its row and column.

TABLE 9 Discriminant validity assessment via heterotrait-monotrait (HTMT) ratio (N = 450).

Construct	(1)	(2)	(3)	(4)	(5)	(6)
(1) Interest/Enjoyment	—					
(2) Perceived Competence	0.013	—				
(3) Effort	0.075	0.101	—			
(4) Pressure/Tension	0.046	0.077	0.254	—		
(5) Perceived Choice	0.090	0.067	0.184	0.433	—	
(6) Value/Usefulness	0.119	0.017	0.165	0.370	0.367	—

HTMT, Heterotrait-Monotrait ratio. Values below the conservative threshold of 0.85 indicate strong discriminant validity (Henseler et al., 2015). All HTMT values in this study are well below 0.85.

TABLE 10 Results of mixed-effects model analysis for intervention effects on total intrinsic motivation scores (n = 600).

Analysis	Statistic	Value	*p*	Effect size [95% CI]
Baseline Equivalence (Pre-test)	*t*(598)	0.97	0.335	Cohen's *d* = 0.08 [−0.08, 0.24]
Mixed-Effects Model				
Group × Time Interaction	F(1,598)F(1,598)	510.12	< 0.001	Partial η^2 = 0.460
Between-Group Comparison (Post-test)				
Experimental vs. Control	*t*(598)	21.34	< 0.001	Cohen's *d* = 1.74 [1.55, 1.93]
Within-Group Changes				
Experimental Group (Pre to Post)	–	–	< 0.001	Cohen's *d* = 1.87 [1.68, 2.06]
Control Group (Pre to Post)	–	–	0.920	Cohen's *d* = −0.01 [−0.20, 0.18]

Intrinsic motivation was measured using the 30-item CIMLQ (Total score range: 30–150). The mixed-effects model included a random intercept for student ID. Effect sizes are reported with 95% confidence intervals.

TABLE 11 Comparison of final examination scores between experimental and control groups (n = 600).

Group	*n*	M	SD	*t*(598)	*p*	Cohen's *d* [95% CI]
Experimental	300	80.10	12.17	3.40	0.001	0.28 [0.12, 0.44]
Control	300	76.57	13.31			

The final examination was a common summative, criterion-referenced assessment, scored by instructors blinded to group allocation.

3.3 Intervention effects on academic performance

The experimental group demonstrated significant improvement in academic performance. An independent-samples t-test revealed that their final exam scores were significantly higher than those of the control group, *t*(598) = 3.40, *p* = 0.001, Cohen's *d* = 0.28. This represents a mean increase of 3.54 points, or a 4.6% improvement relative to the control group's baseline. The smaller standard deviation observed in the experimental group suggests a more concentrated distribution of scores post-intervention. Descriptive statistics and the effect size with its confidence interval are presented in Table 11.

The final examination was a summative, criterion-referenced assessment. Internal consistency reliability was not calculated as the exam was designed for achievement testing rather than psychometric validation. The smaller standard deviation observed in the experimental group (12.17 vs. 13.31 in the control group) alongside a higher mean score suggests more uniform improvement across participants rather than range restriction.

4 Discussion

Based on Self-Determination Theory, the culturally adapted CIMLQ has proven to be a psychometrically valid tool for assessing intrinsic motivation among Chinese engineering students. The intervention grounded in its dimensions yielded significant improvements in both motivation and academic performance. These results can be productively interpreted through an integrated, cross-cultural lens that aligns with recent comparative scholarship.

The intervention's efficacy appears to stem from its sensitivity to both existing cultural strengths and targeted motivational shifts. Many Chinese engineering students enter with well-developed competencies in organization and sustained effort—learning dispositions that align with patterns observed in comparative studies of Hong Kong students (Lo et al., 2024). Components like progress monitoring and goal specification likely amplified these existing proficiencies, directly enhancing Perceived Competence. Concurrently, activities such as reflective sharing and autonomous goal-setting may have facilitated a valuable motivational development. For students with an initial orientation toward achievement and competition—a profile noted in Hong Kong contexts (Lo, 2024)—these practices provided a structured pathway to experience Interest/Enjoyment and Perceived Choice, thereby reducing Pressure/Tension. The embedded, low-friction feedback cycles (goal cards, weekly prompts) served as the operational mechanism, making support for autonomy and competence tangible without overwhelming students—a design principle supported by research on motivation-sensitive feedback (Yang and Chan, 2023).

Regarding measurement methods, this study primarily relied on students' self-report data, which has limitations such as social desirability bias and common method bias.

In terms of testing intervention effectiveness, we found that while short-term incentives could boost students' learning engagement to some extent, the effects were often unsustainable if genuine goal internalization was not achieved. The addition of a "goal internalization" coaching session in the later intervention phase, such as organizing group discussions on the personal meaning of learning content, effectively facilitated the internalization of external motivation. This finding suggests that future educational interventions should not only focus on applying superficial strategies but also emphasize creating a supportive environment conducive to self-integration and meaning construction.

Furthermore, the measurement invariance of the CIMLQ across key subgroups (e.g., gender, academic year) was not tested in this study. Although the scale demonstrated good psychometric properties in the overall sample, the absence of invariance tests limits the certainty with which latent mean comparisons can be made between such groups. Establishing configural, metric, and scalar invariance is an essential next step for future research to confirm that the CIMLQ measures the same constructs in the same way across different populations, thereby strengthening its utility for broader comparative purposes.

The issues discussed above indicate that revising and developing a college student intrinsic motivation questionnaire is a long-term endeavor requiring guidance from scientific theory as well as attention to educational practice. Furthermore, the long-term efficacy and generalizability of educational interventions need to be verified through multiple rounds of iterative experiments. In the future, based on expanding the sample size and extending the tracking period, we will actively integrate multimodal data and intelligent analysis technologies to transition the questionnaire from static assessment to dynamic diagnosis.

5 Conclusion

This study developed the CIMLQ by adapting internationally validated scales, comprising six dimensions: Interest/Enjoyment, Perceived Competence, Effort, Pressure/Tension, Perceived Choice, and Value/Usefulness. Empirical verification established good reliability and validity, supporting its use for assessing intrinsic motivation among Chinese engineering students.

A 19-week teaching experiment demonstrated that structured educational interventions targeting these dimensions significantly enhanced intrinsic motivation levels (21.8% improvement) and substantially improved academic performance.

These findings not only empirically confirm the effectiveness of intrinsic motivation stimulation in higher education but also provide practical approaches for fostering self-directed learning and motivation among engineering students in artificial intelligence contexts.

However, this study presents limitations. The sample was drawn solely from engineering programs at a single university, limiting

generalizability. Although supplemented by behavioral logs, primary reliance on self-report data risks common method bias. Building on the cross-cultural perspective integrated in this study, future research should pursue a specific, testable agenda: (1) Conduct multi-site trials to establish the measurement invariance of the CIMLQ across diverse subgroups and cultural contexts, and verify the intervention's robustness; (2) Experimentally compare different feedback modalities (e.g., integrating blended micro-feedback) within the intervention framework to refine its active components, as suggested by research on feedback design and motivation (Yang and Chan, 2023); (3) Incorporate light application tasks aligned with engineering practice to further strengthen perceived competence and value without increasing cognitive load. This progression would translate the present contributions into a more generalizable, theory-informed model for fostering intrinsic motivation across diverse educational settings.

In summary, this study provides initial empirical foundation and methodological support for theoretical conceptualization and educational intervention of intrinsic motivation among Chinese university students. Future large-scale, multidisciplinary longitudinal research should optimize questionnaire structures and intervention strategies, ultimately aiming to develop systematic and adaptable intrinsic motivation cultivation systems to inform higher education reform and student development.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary material, further inquiries can be directed to the corresponding author.

Ethics statement

This study involved minimal-risk educational practice. Although formal ethics approval was not obtained, it was conducted in strict accordance with the principles of the Declaration of Helsinki, and informed consent was acquired from all participants. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

KY: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Writing – original draft. DH: Supervision, Validation, Project administration, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/feduc.2026.1733781/full#supplementary-material

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